



Original Article

Characteristics of Immune Checkpoint Inhibitor–Related Hepatotoxicity Based on the Baseline Liver Function

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Purpose This study estimated the incidence of immune checkpoint inhibitor–related hepatotoxicity (ICH), identified risk factors, and characterized patients who developed ICH.

Materials and Methods Adult patients treated with immune checkpoint inhibitors (ICIs) from January 2015 to June 2022 in a tertiary hospital were included, excluding those without liver function tests or those with liver cancer but normal baseline liver function. Patients were stratified by baseline liver function status; in overall and each of stratified cohorts ICH incidence was calculated as the number of events per 100 person-years with grade 3 hepatotoxicity as the primary outcome. Patient characteristics were assessed using descriptive statistics, and risk factors were identified through multivariable Cox regression. Causality between ICI use and hepatotoxicity was assessed using the Naranjo Algorithm.

Results Among 803 patients, the ICH incidence was 19.5 cases per 100 person-years, with a higher incidence (47.3 vs. 9.3 cases per 100 person-years) and earlier onset (13 vs. 15 days) in the abnormal compared to the normal group. Significant risk factors for ICH included female sex in the normal group and liver cancer in the abnormal group. According to the Naranjo Algorithm, all the 60 ICH cases were classified as “probable” or “possible”. Among the 60 cases, 61.7% (n=37) resulted in ICI discontinuation. The baseline liver function did not impact on the severity or the likelihood of ICI discontinuation.

Conclusion Future studies are needed to evaluate whether the impact of ICI discontinuation on survival outcomes in patients with ICH varies based on baseline liver function abnormalities.

Key words Immune checkpoint inhibitors, Hepatotoxicity, Causality assessment

Introduction

The advancement of immune checkpoint inhibitors (ICIs) in cancer therapy over the last 10 years has led to their integration into the standard-of-care for various cancer types [1]. The broad use of ICIs has yielded substantial real-world data on their safety and effectiveness [2-6]. ICIs block the interaction between immune checkpoints like programmed death-1 (PD-1) and cytotoxic T lymphocyte antigen-4 (CTLA-4) with their ligands, or directly inhibit a specific ligand called programmed death-ligand 1 (PD-L1). The inhibition of immune checkpoints bolsters the cancer patient's innate immune response against cancer cells [7]. While ICIs have significantly improved survival outcomes, they have also been associ-

ated with an increase in the occurrence of immune-related adverse events (irAEs).

The immune mechanisms underlying ICIs suggest a theoretical rationale for associating the development of irAEs with improved survival in ICI-treated cancer patients, as activation of the immune system can lead to both tumor responses and autoimmunity [8-10]. However, the presence of irAEs does not consistently predict favorable outcomes. The clinical manifestations of irAEs are relatively obscure and not straightforward to assert as indicator of beneficial treatment progress [11]. Severe irAEs can often lead to deterioration of organ functions, particularly in the organs where the cancer originated, thereby increasing morbidity, health-care cost, and mortality among cancer patients [11].

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Hepatotoxicity including hepatitis and cholangitis is known to occur in up to 23% of ICI-treated patients in clinical trials and observational studies [12-15]. Liver cancer, such as hepatocellular carcinoma, and other liver disease are well-known risk factors for ICI-associated hepatotoxicity (ICH). Several studies have reported that the incidence of hepatotoxicity varies by type and dose of ICIs, with the highest risk observed in treatments involving CTLA-4 inhibitors, particularly when combined with PD-1 inhibitors or used in high doses [16-21].

A recent meta-analysis found that ICI treatment significantly enhances survival among Asian cancer patients, demonstrating better outcomes compared to non-Asian patients treated with PD-1/PD-L1 inhibitors, while immune-related safety studies among Asian population are limited [22]. Characterizing hepatotoxicity in Asian patients will add valuable insights to optimize treatment strategies and manage drug-induced liver failure during ICI treatment. To improve our understanding of ICI-use related hepatotoxicity, we aim to estimate the incidence of hepatotoxicity and identify risk factors among Korean patients initiating ICIs, as well as characterize the patients who developed hepatotoxicity.

Materials and Methods

1. Data source

We used the electronic health records at Pusan National University Hospital (PNUH) converted to the Observational Medical Outcomes Partnership (OMOP) - Common Data Model (CDM) [23]. This dataset encompasses patient demographics, diagnoses, drug administration/prescriptions, and laboratory test orders and results for patients visiting the hospital between January 2015 and June 2022.

PNUH employs the Korean Standard Classification of Disease and Causes of Death, version 7 (KCD-7) as the medical diagnosis coding system. The KCD-7 is the Korean adaptation of the International Classification of Diseases, the 10th version (ICD-10) and has been mapped to SNOMED-CT within the CDM at PNUH. Laboratory test types and results were mapped to Logical Observation Identifiers Names and Codes (LOINC), an international standard for coding medical laboratory observations; and medications were mapped using RxNorm or RxNorm Extension within the CDM at PNUH. RxNorm provides a standardized nomenclature for clinical drugs used in the United States, while RxNorm Extension covers drugs that used exclusively outside the U.S. SNOMED-CT, LOINC, RxNorm, and RxNorm Extension are all standardized vocabularies within the OMOP framework (S1 and S2 Tables).

2. Study design

We conducted a retrospective cohort study of patients aged 18 or older who received at least one dose of ICI at PNUH between January 2015 and June 2022. The ICIs of interest included CTLA4 inhibitors (ipilimumab), PD-1 inhibitors (cemiplimab, dostarlimab, nivolumab, and pembrolizumab), and PD-L1 inhibitors (atezolizumab, avelumab, and durvalumab). Patients were excluded if there were no liver function test results recorded within 3 months before the first ICI administration (=the index date). To assess the impact of baseline liver function on hepatotoxicity, we reviewed aspartate aminotransferase (AST), alanine transaminase (ALT), and total bilirubin (T-Bil) measured within 90 days prior to or on index date. Normal liver function was defined according to the institutional reference ranges of Pusan National University Hospital: AST $\leq$ 40 IU/L, ALT $\leq$ 40 IU/L, and T-Bil $\leq$ 1.2 mg/dL. Patients were classified as having nor-

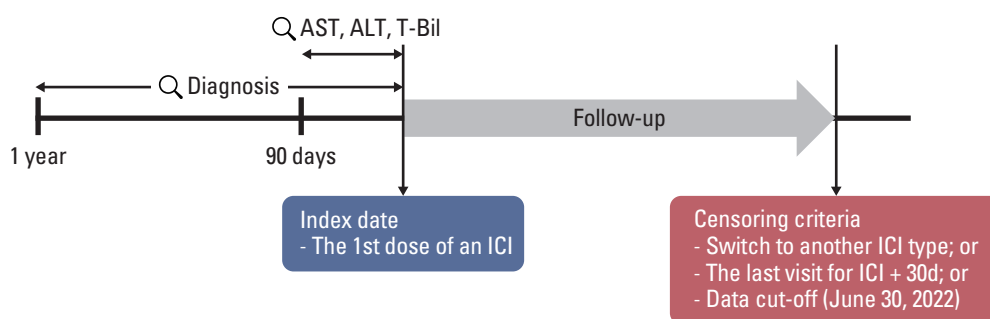


Fig. 1. Study design. In order to classify patients between normal and abnormal group, we observed liver function tests results (aspartate aminotransferase [AST], alanine aminotransferase [ALT], and total bilirubin [T-Bil]) up to 90 days prior to the index date (=the first dose of an immune checkpoint inhibitor), and for liver cancer diagnoses up to 1 year before the index date. During the 1 year, we also observed baseline characteristics for study population. The follow-up period started from the index date to the time of switching to another immune checkpoint inhibitors (ICI) type, 30 days after the last ICI dose, or the data cut-off (June 30, 2022), whichever came first.

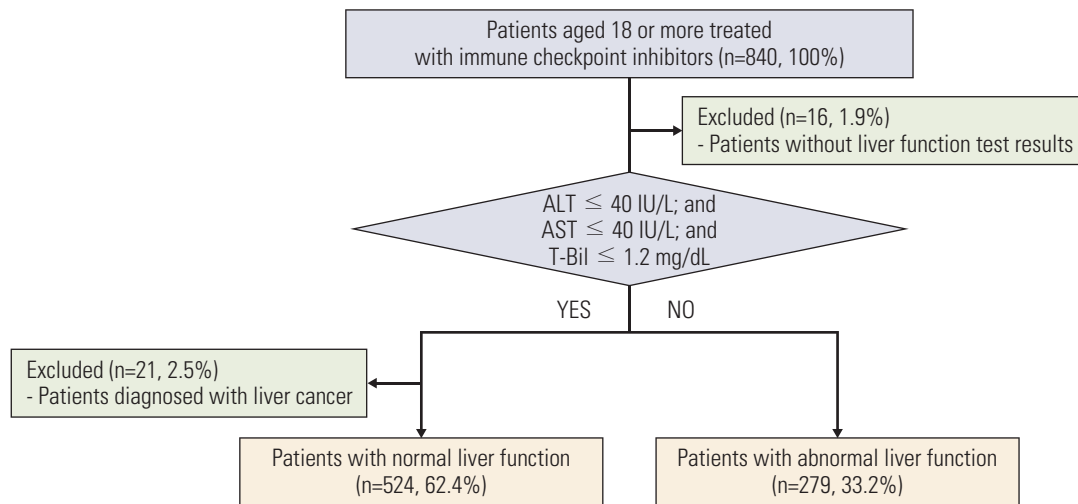


Fig. 2. Flow diagram for study population. ALT, alanine aminotransferase; AST, aspartate aminotransferase; liver cancer, malignant neoplasm of liver and intrahepatic bile ducts; T-Bil, total bilirubin.

mal liver function only if all values met the following criteria throughout the 90-day period: Any value exceeding the cut-off resulted in classification as abnormal liver function. Among the normal group, patients were further excluded if they had a history of liver cancer, confirmed by having KCD-7 of C22 (Malignant neoplasm of liver and intrahepatic bile ducts) within up to 365 days prior to the index date. Of note, patients with a history of hepatitis or other chronic liver diseases were eligible for inclusion in the normal group, provided they exhibited no abnormal liver function during the 90-day period prior to the index date. Censoring occurred at switching to another ICI type, 30 days after the last ICI administration or at the end of data availability (June 30, 2022) whichever came first (Fig. 1).

3. Outcome definition

The outcome definition was adapted from grade 3 or higher hepatotoxicity as defined by the Common Terminology Criteria for Adverse Events (CTCAE) ver. 5.0 [24]. The primary endpoint was AST or ALT greater than 5 times of either (1) upper limit normal (ULN), if baseline was normal; or (2) baseline, if baseline liver function was abnormal; or T-Bil greater than 3 times of ULN (or baseline) if baseline was normal (or abnormal).

4. Causality assessment

To assess hepatotoxicity causality, we developed detailed visual presentations to facilitate the application of the Naranjo Algorithm [25], a standardized tool for evaluating adverse drug reactions. The figures included several key components: the index date (i.e., the date of first ICI administra-

tion), the outcome (grade 3 or higher hepatotoxicity) date, and, between these two points, the date of CTCAE version 5.0 hepatotoxicity grade 1 or 2 based on liver function test results using variables available in the CDM.

Additionally, the figures included the date and type of all drugs administered or prescribed during the follow-up. Among these, hepatotoxic drugs identified from the LiverTox database, a resource from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) were highlighted [26]. LiverTox classifies drugs into category A or B hepatotoxic agents; both category A and B drugs are known to cause idiosyncratic liver injury, with category A drugs posing a greater risk than category B drugs.

Using these figures, two independent reviewers applied the Naranjo algorithm to conduct causality assessments for identified hepatotoxicity cases. Any discrepancies were resolved by discussions with a third reviewer. All cases were classified into four levels according to the likelihood of causality between ICI use and hepatotoxicity: doubtful, possible, probable, or definite.

5. Statistical analysis

Patient characteristics were summarized with descriptive statistics and compared between normal and abnormal groups using t test or chi-square test with Fisher's correction where appropriate. Hepatotoxicity incidence was estimated by events per 100 person-years in the overall and stratified groups by (1) the use of chemotherapy during follow-up and (2) baseline liver function. Multivariable Cox regressions were used to identify risk factors for ICH. Patient age, sex, type of ICI, cancer types (liver cancer, other major cancers,

Table 1. Characteristics of study population

Characteristic	Total (n=803)	Normal ^{a)} (n=524)	Abnormal ^{b)} (n=279)	p-value
At index date				
Sex				
Male	563 (70.1)	354 (67.6)	209 (74.9)	0.030
Female	240 (29.9)	170 (32.4)	70 (25.1)	
Age (yr)	67.1±10.1	68.3±9.5	64.8±10.7	< 0.001
Type of ICI at initiation				
Atezolizumab	275 (34.2)	170 (32.4)	105 (37.6)	0.140
Pembrolizumab	264 (32.9)	188 (35.9)	76 (27.2)	0.013
Nivolumab	215 (26.8)	129 (24.6)	86 (30.8)	0.059
Durvalumab	29 (3.6)	21 (4.0)	8 (2.9)	0.410
Ipilimumab+nivolumab	20 (2.5)	16 (3.1)	4 (1.4)	0.161
Baseline period				
Liver function test				
ALT	25.5±36.7	14.4±6.6	46.3±55.9	< 0.001
AST	33.8±42.7	19.7±5.8	60.4±64.1	< 0.001
Total bilirubin	0.5±0.7	0.4±0.2	0.9±1.1	< 0.001
Cancer type				
Lung	445 (55.4)	320 (61.1)	125 (44.8)	< 0.001
Liver	77 (9.6)	-	77 (27.6)	< 0.001
Stomach	42 (5.2)	23 (4.4)	19 (6.8)	0.142
Gallbladder and bile duct	18 (2.2)	10 (1.9)	8 (2.9)	0.382
Colon	16 (2.0)	9 (1.7)	7 (2.5)	0.445
Lymphoma	13 (1.6)	7 (1.3)	6 (2.2)	0.391
Prostate	10 (1.2)	7 (1.3)	3 (1.1)	1.000
Breast	6 (0.7)	5 (1.0)	1 (0.4)	0.671
Leukemia	3 (0.4)	1 (0.2)	2 (0.7)	0.278
Pancreas	3 (0.4)	1 (0.2)	2 (0.7)	0.278
Other cancers ^{c)}	206 (25.7)	158 (30.2)	48 (17.2)	< 0.001
Chronic liver disease				
Hepatitis	64 (8.0)	7 (1.3)	57 (20.4)	< 0.001
Other liver diseases ^{d)}	91 (11.3)	15 (2.9)	76 (27.2)	< 0.001
Follow-up period				
No. of ICI cycles				
1	138 (17.2)	70 (13.4)	68 (24.4)	< 0.001
2	121 (15.1)	56 (10.7)	65 (23.3)	
3	130 (16.2)	94 (17.9)	36 (12.9)	
4	90 (11.2)	67 (12.8)	23 (8.2)	
5 or more	324 (40.3)	237 (45.2)	87 (31.2)	

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and others), and liver disease types (hepatitis, others) were included in the model as potential risk factors. The details of each category for cancer types and liver disease are available in S2 Table. Results were considered statistically significant at an alpha level of < 0.05. All statistical analyses were performed using SAS version 9.4 (SAS Institute Inc.).

Results

Among 840 ICI users, a total of 803 patients were selected according to the inclusion and exclusion criteria (Fig. 2). A total of 524 patients (62.4%) were classified as the normal group (i.e., having normal liver function at ICI initiation) and the remainder (n=279, 37.6%) was classified as the abnormal group. Patient characteristics are shown in Table 1. The average age was 67.1 years (standard deviation, 10.1) and more

Table 1. Continued

Characteristic	Total (n=803)	Normal ^{a)} (n=524)	Abnormal ^{b)} (n=279)	p-value
Chemotherapy				
No	544 (67.7)	340 (64.9)	204 (73.1)	< 0.001
Yes	259 (32.3)	184 (35.1)	75 (26.9)	
Carboplatin	141 (17.6)	99 (18.9)	42 (15.1)	0.173
Cisplatin	41 (5.1)	30 (5.7)	11 (3.9)	0.275
Etoposide	57 (7.1)	44 (8.4)	13 (4.7)	0.050
Gemcitabine	45 (5.6)	30 (5.7)	15 (5.4)	0.838
Pemetrexed	43 (5.4)	31 (5.9)	12 (4.3)	0.333
Others ^{c)}	144 (17.9)	103 (19.7)	41 (14.7)	0.081

Values are presented as number (%) or mean±SD. ALT, alanine aminotransferase; AST, aspartate aminotransferase; ICI, immune checkpoint inhibitor; SD, standard deviation. ^{a)}Normal: defined by AST and ALT levels ≤ 40 IU/L and total bilirubin ≤ 1.2 mg/dL, ^{b)}Abnormal: defined by AST or ALT > 40 IU/L or total bilirubin > 1.2 mg/dL, ^{c)}Other cancers: all other types of cancer not listed in the table including but not limited to Malignant melanoma of skin (C43), Malignant neoplasm of cervix uteri (C53), Malignant neoplasm of ovary (C56), Malignant neoplasm of kidney, except renal pelvis (C64), Malignant neoplasm of bladder (C67), ^{d)}Other liver diseases: Alcoholic liver disease (K70); Toxic liver disease (K71); Hepatic failure, not elsewhere classified (K72); Chronic hepatitis, not elsewhere classified (K73); Fibrosis and cirrhosis of liver (K74); Other inflammatory liver diseases (K75); Other diseases of liver (K76); Liver disorders in disease classified elsewhere (K77), ^{e)}Others: docetaxel, irinotecan, paclitaxel, dacarbazine, oxaliplatin, capecitabine, 5-fluorouracil, doxorubicin, topotecan, ifosfamide, methotrexate, cyclophosphamide, vinorelbine, cytarabine, mitomycin, tegafur, mitoxantrone, vinblastine, vincristine, gefitinib, eribulin.

than 70% of the study population (n=563) were male patients. Atezolizumab (n=275, 34.2%), pembrolizumab (n=264, 32.9%), and nivolumab (n=215, 26.8%) were the most commonly used ICIs. Lung cancer (n=445, 55.4%) was the most common cancer type followed by liver cancer (n=77, 9.6%).

During the follow-up, patients received a median of 4 ICI cycles (interquartile range [IQR], 2 to 7). A total of 259 (32.3%) received subsequent chemotherapy, with carboplatin being the most common (n=141, 17.6%) (S3 Table). Significant differences were noted between the groups in terms of baseline characteristics. The normal group was older (68.3 vs. 64.8 years), had more female patients (32.4% vs. 25.1%), and more frequently received pembrolizumab (35.9% vs. 27.2%), or a nivolumab and ipilimumab combination (3.1% vs. 1.4%). They also received more ICI cycles (median, 4 vs. 3).

The overall hepatotoxicity incidence was 19.5 cases per 100 person-years, and the incidence was 5 times higher in the abnormal group (47.3 vs. 9.3 cases/100 person-years). The incidence was lower in patients who received chemotherapy during follow-up compared to those who did not, with the highest hepatotoxicity incidence (57.9 cases/100 person-years) observed in patients with abnormal baseline liver function without concomitant chemotherapy during follow-up (Table 2).

Table 3 presents the results from multivariable Cox regression analyses aimed at identifying risk factors for ICH within stratified groups based on baseline liver function. In the normal group, female sex was significantly associated with the development of ICH with an adjusted hazard ratio (aHR) of

2.66 with a 95% confidence interval (CI) of 1.05-6.72. In the abnormal group, liver cancer was significant risk factor for ICH (aHR, 3.67; 95% CI, 1.16 to 11.64).

Table 4 summarizes clinical characteristics of hepatotoxicity. Of the 60 identified hepatotoxicity cases, five cases (8.3%) were classified as “probable”, while the remaining cases were considered “possible” ICH cases according to the Naranjo Algorithm assessment. Grade 3 or greater hepatotoxicity was identified a median of 41.5 days (IQR, 15 to 101.5) following the first ICI dose, typically after a median of 2 doses. The median interval from the last ICI dose to the onset of hepatotoxicity was approximately two weeks.

By 60 days after ICH detection, 39 patients (65.0%) had not resumed ICI treatment while the remaining 21 patients received one (n=8, 38.1%), two (n=6, 28.6%), or three (n=7, 33.3%) additional doses. Patients with abnormal liver function experienced hepatotoxicity earlier (median, 32 days vs. 57 days) and had a slightly higher post-ICH mortality rate compared to those with normal liver function (35.9% vs. 28.6%), but the difference was not statistically significant (p=0.566). The severity and rate of discontinuation of ICI are comparable between both groups.

Fig. 3A and B illustrate example cases of “probable” hepatotoxicity following the second dose of atezolizumab, representing the highest-rated cases in the normal and abnormal groups, respectively. The remainder of 60 ICH cases are illustrated in S4 Fig.

Table 2. Incidence of hepatotoxicity among ICI users stratified by baseline liver function

	Total		Normal group		Abnormal group	
	Incidence (%) (event/at risk)	Incidence (event/100 py)	Incidence (%) (event/at risk)	Incidence (event/100 py)	Incidence (%) (event/at risk)	Incidence (event/100 py)
Total	7.5 (60/803)	19.5	4.0 (21/524)	9.3	14.0 (39/279)	47.3
Concomitant chemotherapy during follow-up						
No	9.2 (50/544)	23.5	4.4 (15/340)	9.8	17.2 (35/204)	57.9
Yes	3.9 (10/259)	10.6	3.3 (6/184)	8.3	5.3 (4/75)	18.1

ICI, immune checkpoint inhibitor; py, person-years.

Table 3. Risk factors for ICIs related hepatotoxicity by baseline liver function

Variable	Normal		Abnormal	
	Crude HR	aHR ^{a)} (95% CI)	Crude HR	aHR ^{a)} (95% CI)
Characteristic				
Age	1.01 (0.96-1.05)	1.02 (0.97-1.06)	1.01 (0.98-1.04)	1.01 (0.98-1.04)
Sex				
Male		Reference		
Female	2.48 (1.05-5.86)	2.66 (1.05-6.72)	1.09 (0.53-2.25)	1.06 (0.50-2.23)
Type of ICI at initiation				
Anti-PD-L1		Reference		
Anti-PD-1	2.54 (0.86-7.56)	2.21(0.73-6.65)	1.09 (0.57-2.07)	1.95(0.96-3.98)
Cancer type				
Other cancers ^{b)}		Reference		
Liver cancer	N / A	N / A	3.25 (1.23-8.64)	3.67 (1.16-11.64)
Other major cancer ^{c)}	1.1 (0.42-2.83)	1.42 (0.53-3.83)	0.78 (0.28-2.18)	0.78 (0.27-2.26)
Chronic liver disease^{d)}				
No		Reference		
Yes	1.49 (0.20-11.19)	1.42 (0.19-10.71)	2.68 (1.42-5.03)	1.18 (0.48-2.92)

CI, confidence interval; HR, hazard ratio; ICI, immune checkpoint inhibitor; N / A, not available; PD-1, programmed death-1; PD-L1, programmed death-ligand 1. ^{a)}aHR: adjusted hazard ratio; all variables were mutually adjusted in a Cox regression model, ^{b)}Other cancers: bladder, cervix uteri, kidney, ovary, skin, etc., ^{c)}Other major cancer: lung, stomach, gallbladder and bile duct, colon, lymphoma, prostate, breast, leukemia, pancreas, ^{d)}Chronic liver disease: Alcoholic liver disease (K70); Toxic liver disease (K71); Hepatic failure, not elsewhere classified (K72); Chronic hepatitis, not elsewhere classified (K73); Fibrosis and cirrhosis of liver (K74); Other inflammatory liver diseases (K75); Other diseases of liver (K76); Liver disorders in disease classified elsewhere (K77); Acute hepatitis A (B15); Acute hepatitis B (B16); Other acute viral hepatitis (B17); Chronic viral hepatitis (B18); Unspecified viral hepatitis (B19).

Discussion

Our study aimed to estimate the incidence of ICH, identify its risk factors, and characterize the clinical profiles of patients who developed hepatotoxicity among ICI-treated cancer patients in a tertiary hospital in Korea. The incidence of hepatotoxicity in the overall cohort (7.5%), the normal liver function group (4.0%), and the abnormal liver function group (14.0%) all fell within the previously reported range of 0.7% to 16% according to a recent systematic review [12], suggesting that Asian ethnicity may not influence the fre-

quency of ICH development during ICI treatment. Notably, patients with pre-existing liver dysfunction experienced ICH more frequently and had an earlier onset compared to those with normal baseline liver function. However, we found that the severity of ICH and the rate of ICI discontinuation did not significantly differ between the two groups.

Our findings are consistent with previous studies conducted in Asian populations. For instance, Ito et al. (2021) [27] reported a 4.8% incidence of grade ≥ 3 hepatotoxicity in Japanese patients who did not have liver cancer or chronic liver disease. These criteria are comparable to those used for

Table 4. Characteristics of 60 hepatotoxicity cases

Characteristic	Total (n=60)	Normal ^{a)} (n=21)	Abnormal ^{a)} (n=39)	p-value
Days from the first ICI dose	41.5 (15-101.5)	57 (38-105)	32 (13-98)	0.092
Days from the last ICI dose	13 (8-20.5)	15 (10-24)	13 (5-16)	0.210
No. of ICI administration	2 (1.5-4)	3 (2-4)	2 (1-4)	0.308
Causality assessment				
Probable	5 (8.3)	1 (4.8)	4 (10.3)	0.649
Possible	55 (91.7)	20 (95.2)	35 (89.7)	
Hepatotoxicity severity				
Grade 3	54 (90.0)	18 (85.7)	36 (92.3)	0.655
Grade 4 or higher	6 (10.0)	3 (14.3)	3 (7.7)	
Results of hepatotoxicity				
ICI continued ^{b)}	21 (35.0)	7 (33.3)	14 (35.9)	0.843
Death	2 (3.3)	1 (4.7)	1 (2.5)	> 0.99
ICI discontinued ^{c)}	37 (61.7)	14 (66.7)	23 (59.0)	0.559
Death	18 (30.0)	5 (23.8)	13 (33.3)	0.443
ICI resumed ^{c)}	1 (1.7)	1 (4.7)	0	0.350
Lost to follow-up	2 (3.3)	0	2 (5.1)	0.537

Values are presented as median (IQR) or number (%). ALT, alanine aminotransferase; AST, aspartate aminotransferase; ICI, immune checkpoint inhibitor; IQR, interquartile range. ^{a)}Normal was defined by AST and ALT levels ≤ 40 IU/L and total bilirubin ≤ 1.2 mg/dL; and abnormal was defined by AST or ALT > 40 IU/L or total bilirubin > 1.2 mg/dL. ^{b)}ICI continued was defined as the presence of an ICI administration record within 60 days after the onset of hepatotoxicity. ^{c)}ICI discontinued was defined as the absence of an ICI administration record within 60 days after the onset of hepatotoxicity. Among these patients, those who subsequently restarted ICI therapy beyond the 60-day window were classified as having resumed treatment.

our normal liver function group, which showed a similar incidence of 4.0%. Jiang et al. (2024) [28] also identified liver cancer and chronic hepatitis B virus infection as significant risk factors for ICI-related liver injury among Chinese ICI users. This aligns with our observation that liver cancer was a major risk factor in patients with abnormal liver function. In a recent Korean study, Park et al. (2025) [29] reported a 22.5% incidence of hepatotoxicity and noted the predominance of cholestatic and mixed-type liver injuries, highlighting the importance of phenotype-specific clinical strategies. Compared to these studies, our study provides additional value by stratifying patients according to baseline liver function and by applying a structured causality assessment using the Naranjo Algorithm.

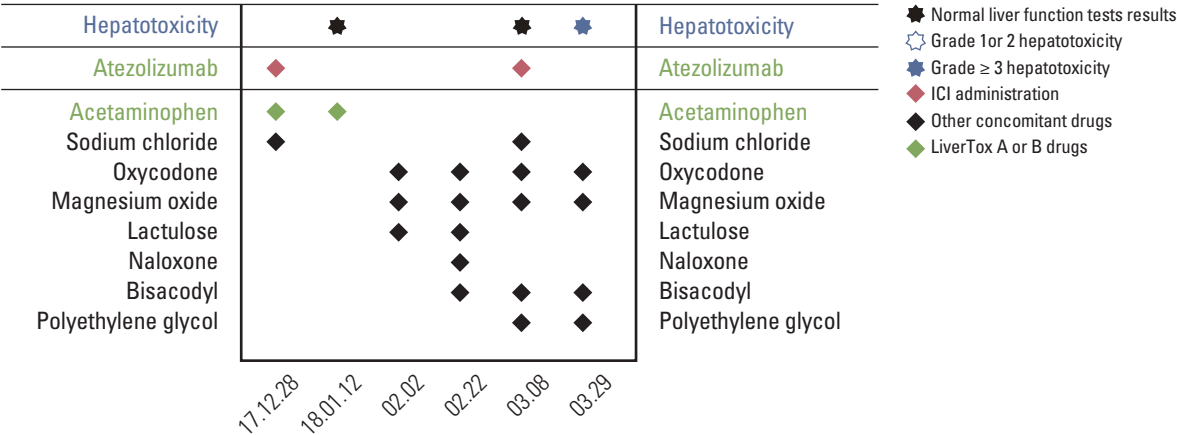
These findings add to a growing body of evidence suggesting that baseline liver function and comorbid conditions influence the clinical course and outcomes of immune-related adverse events among ICI-treated patients [30]. Specifically, our study findings observed that ICH patterns varied by liver function status, with implications for phenotype-tailored management to optimize treatment outcomes after ICH development. Future studies with larger sample sizes are needed to adequately assess the impact of ICH development on mortality differences between groups, considering the presence or absence of ICI discontinuation.

According to the American Society of Clinical Oncology guidelines, grade 3 or higher hepatotoxicity warrants holding ICI therapy, initiating corticosteroids, and strongly considering permanent discontinuation [31]. In this study population, 35% (21/60) of patients who experienced hepatotoxicity continued ICI therapy. However, when 14 patients who ultimately discontinued ICI after receiving one or two additional doses are reclassified, the proportion of patients considered to have continued therapy decreases to 11.7% (7/60). All of these patients had available liver function test results, and most had returned to normal liver function, except for 4 cases. Although the specific reasons for continuing ICI therapy in these patients could not be determined from the available data, it is clinically plausible that their hepatotoxicity was not attributed to irAEs.

Our study observed a relatively lower incidence of hepatotoxicity in patients receiving ICIs in combination with chemotherapy. However, this finding should be interpreted with caution. Given that most chemotherapeutic agents themselves carry a risk of hepatotoxicity, it is not plausible to suggest that chemotherapy confers a protective effect against ICI-induced hepatotoxicity. Moreover, combination regimens are typically selected based on tumor type and standard treatment protocols, rather than individual patients' baseline health status. Therefore, the observed low-

A

Case example 1. Age: 86 / Sex: M / Cancer type: others
Naranjo score: 6 / Naranjo algorithm: probable



B

Case example 2. Age: 67 / Sex: M / Cancer type: liver cancer, hepatitis or other liver disease
Naranjo score: 7 / Naranjo algorithm: probable

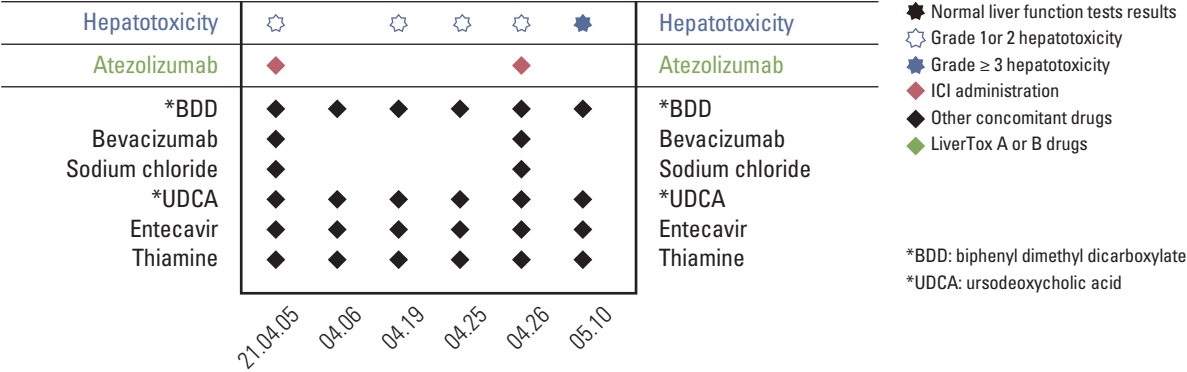


Fig. 3. Temporality assessment for immune checkpoint inhibitor (ICI)–related hepatotoxicity causality. (A) A probable case of ICI-related hepatotoxicity in an 86-year-old male patient following two cycles of atezolizumab. The patient had normal liver function prior to initiating ICI therapy. Liver function remained normal during two live function tests after the first dose. No alternative causes were identified for the subsequent hepatotoxicity, implicating atezolizumab as the likely cause. (B) Another probable ICI-related hepatotoxicity case in a 67-year-old male patient with pre-existing liver disease. At the time of atezolizumab initiation, the patient had grade 1 hepatotoxicity, which progressed to grade 3 hepatotoxicity after the second dose, with no other alternative causes identified.

er incidence may reflect differences in clinical context rather than a true protective effect. For instance, patients receiving combination therapy may have been treated in the first-line setting, which could indicate a relatively better overall health status at treatment initiation. Further investigation, including analyses stratified by treatment line and clinical setting, is warranted to better understand this association and appropriately contextualize the safety profile of ICI-chemotherapy combinations.

Our study has several strengths. First, we conducted a series of causality assessments using the Naranjo algorithm–

a method often overlooked in large cohort studies evaluating safety outcome. By creating visual aids that include essential components of causality assessment, particularly for hepatotoxicity events, we streamlined the application of the Naranjo algorithm, making it more accessible and systematic. We developed an automated tool to generate these figures in settings where ICI patient cohorts can be constructed using CDM; details of this tool will be summarized elsewhere. Additionally, our use of the OMOP CDM, ensured a high level of data consistency and interoperability, which enhances both the reproducibility and validity of our findings.

This framework allowed for precise data mapping of clinical events and laboratory results, facilitating more accurate identification and comparison of hepatotoxicity incidences across patient groups. Finally, our study is one of the few to focus on an Asian patient population, filling a critical gap in understanding immune-related hepatotoxicity in this demographic. Given the underrepresentation of Asian patients in clinical trials of ICIs, our findings provide valuable insights that may aid in the optimization of hepatotoxicity management strategies tailored to this population. These strengths collectively enhance the clinical relevance and applicability of our findings and contribute meaningful evidence to the literature on ICI safety and efficacy.

Despite its strengths, our study has several limitations. First, due to the inherent limitations of secondary data, we could not fully exclude the possibility that factors other than ICIs may have contributed to hepatotoxicity. To address this, we applied the Naranjo algorithm to each identified case, incorporating available information on concomitant hepatotoxic medications and pre-existing liver disease. All 60 cases were classified as either 'possible' or 'probable' adverse drug reactions. Although liver disease was included as a covariate in the regression model, limited statistical power precluded a more granular analysis of individual risk factors. Furthermore, the relatively small sample size and single-center design may limit the generalizability of our findings, particularly given the unique demographic and clinical characteristics of our cohort. Future studies using larger, multi-center datasets are warranted to validate these findings and enhance their broader applicability. Second, although the Naranjo algorithm provided a structured approach for causality assessment, its reliance on clinical judgment and available data may introduce some degree of subjectivity. Third, using fixed cut-off values for AST, ALT, and T-Bil may misclassify patients with subclinical liver dysfunction as having normal liver function, potentially overestimating hepatotoxicity in the normal group. In our study, 25 such patients (AST or ALT 38-40 IU/L) were included in the normal group, but none developed hepatotoxicity, suggesting minimal impact from this misclassification.

Fourth, the reasons for ICI discontinuation could not be fully ascertained from the available data. Although some of hepatotoxicity cases in this study had temporally coincided with ICI discontinuation, we cannot exclude the possibility that underlying disease progression contributed to liver function deterioration or influenced the clinical decision to withhold treatment. Notably, the absence of radiographic or clinical progression data limits our ability to differentiate between discontinuation due to irAEs and that due to lack of efficacy. Therefore, ICI discontinuation in this study should not be interpreted as having a direct prognostic impact.

Finally, our study focused primarily on baseline liver function and hepatotoxicity outcomes, with limited exploration of long-term survival impacts and detailed causes of death associated with ICH. While we examined the short-term prognostic impact of ICI discontinuation following ICH, further studies with extended follow-up are needed to fully assess the influence of ICH on overall survival and patient outcomes.

In conclusion, our study demonstrates that ICH incidence is substantially higher and occurs earlier in patients with abnormal baseline liver function compared to those with normal liver function. Risk factors for ICH varied by liver function status: female sex was linked to increased risk in the normal group, whereas liver cancer was a prominent risk factor in the abnormal group. Despite differences in time to onset and incidence, the severity of ICH and the likelihood of ICI discontinuation were similar across groups, indicating that baseline liver function may not be a determinant of ICH severity or continuation of ICI therapy due to ICH. These findings highlight the importance of close monitoring for ICH in patients with pre-existing liver impairment and point to the potential benefits of tailored management strategies for those at higher risk. Further studies are needed to understand the long-term survival implications of ICI discontinuation after ICH and to improve overall health outcomes for these patients.

Electronic Supplementary Material

Supplementary materials are available at Cancer Research and Treatment website (<https://www.e-crt.org>).

Ethical Statement

This study was reviewed and exempted by the Institutional Review Board (IRB) of Pusan National University Hospital (IRB No. 2023-0266). The need for informed consent was waived by the IRB due to the retrospective nature of the study.

Author Contributions

Conceived and designed the analysis: Jung WJ, Jo EJ, Kim YJ, Park M, Kim E, Jung YS, Park SR, Oh JS, Jung SY, Jeon N.

Collected the data: Jeon N.

Contributed data or analysis tools: Jo EJ, Kim YJ, Jung YS, Park SR, Oh JS, Kim SH, Park J, Jung SY, Jeon N.

Performed the analysis: Jung WJ, Park M, Kim E, Kim SH, Park J.

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Conflicts of Interest

Conflict of interest relevant to this article was not reported. N Jeon, WJ Jung, J Park and S-H Kim are co-inventors on the pending patent related to this study.

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